

ARYA SANDESH THAKUR

ARTIFICIAL INTELLIGENCE & DATA SCIENCE ENGINEERING STUDENT

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EDUCATION

B.Tech in Artificial Intelligence & Data Science 2025 – 2029 (Expected)
REVA University

Focusing on advanced machine learning algorithms, deep structures, data architectural patterns, and computer statistics metrics.

Higher Secondary Education (11th & 12th) Completed 2025
PACE College

Specialization in Science stream focus areas (Physics, Chemistry, Mathematics).

High Schooling (ICSE) Graduated
St. Mary's ICSE School

Foundation curriculum finishing with strong core academic analytical skills.

CORE PROJECTS

2D Graphics Editor in C

A low-level graphic rendering and editing framework built in C using dynamic pixel-buffer arrays to process user inputs, execute custom vector transformations, and render visual primitives without heavy third-party GUI dependencies.

- **Rasterization Engine:** Implemented custom algorithms (Bresenham's line & midpoint circle) to efficiently draw, stroke, and fill 2D vector shapes directly on a 2D canvas matrix.
- **Memory & Performance:** Optimized dynamic heap allocations and stack operations to manage drawing state histories (undo/redo functionality) with minimal overhead.
- **File I/O Parsing:** Developed custom binary file export and load subroutines to serialize canvas data, palette structures, and image layouts across sessions.

Personal Portfolio Website

A responsive personal developer portfolio engineered to showcase engineering projects, technical credentials, and interactive web tools with optimized performance and accessibility.

- **Frontend Architecture:** Structured modular, semantic HTML5/CSS3 components with responsive layouts optimized for all device breakpoints and cross-browser compatibility.
- **Interactive Canvas Integration:** Embedded custom JavaScript canvas animations and dynamic theme toggles to enhance interactive visual engagement.
- **Deployment & CI/CD:** Implemented continuous deployment workflows via GitHub Actions & GitHub Pages with automated asset minification and performance monitoring.

TECHNICAL TOOLKIT

Python

C Programming

Machine Learning

Data Science

Internet of Things (IoT)

Web Development

Git & GitHub

Data Visualization

HTML5 & CSS3

JavaScript

CREDENTIALS

Machine Learning Specialist

Verified core algorithm architecture metrics.

Advanced Python for Data Science

Validation of structured cleaning & analytics mechanics.